

Ex-Funding Days: How Perpetual Futures Price a Scheduled Payment

Martingale · Research Note 007 Author: Neil Gilani · Code: [pipeline.py](#), [experiment.py](#), [holdout.py](#) · Pre-registration: [377bf67](#) · Addendum: [e51757a](#)

Abstract

Perpetual futures pay funding between longs and shorts at pre-announced timestamps, and only positions open at the timestamp pay or receive it. Frictionless no-arbitrage therefore requires the perpetual's basis to jump by the funding rate at settlement: the ex-dividend price drop, mirrored and two-signed. We test this in a pre-registered design on 431,752 Binance settlements across 62 contracts from 2020 to 2026. **At the timestamp, the basis prices only 6.0% of the payment;** the frictionless value is rejected at $t = -105$. Placebo hours show nothing, and clock hours that become settlement times after a funding-interval change acquire the effect. The reason is institutional: Binance settles over a window of up to a minute, with a ± 15 -second uncertainty about which positions are counted, so there is no single instant to price. **Across the three minutes spanning that window, the basis reprices 44% of the payment, after drifting 42% the other way during the preceding hour.** Both figures were filed as an addendum before any further data were downloaded, and both replicate on 34 contracts never previously examined: **0.449 and -0.422**, on 199,838 settlements. Across the window the perpetual's own price moves by roughly the full payment, and Binance spot moves by about half of it, so the derivative's payment schedule moves the underlying market. A hedged trader who holds through the window keeps 56–59% of large funding payments, about 5–7 bp per event after 8 bp of costs. The predicted post-settlement drift and two joint hypotheses did not replicate.

1. Introduction

Perpetual futures are the dominant cryptocurrency derivative. They never expire, so instead of converging to spot at maturity they exchange a **funding payment** between longs and shorts at fixed timestamps: every eight hours at 00:00, 08:00 and 16:00 UTC for most Binance contracts. When the funding rate F is positive, longs pay shorts $F \times$ notional; when it is negative, shorts pay longs. The rule that matters here is that **only positions open at the settlement timestamp pay or receive the payment.**

That rule makes settlement an unusually clean asset-pricing event. A long who holds through the timestamp pays F ; a long who buys one second later does not. If the price did not change across the timestamp, a trader could short just before a positive-funding settlement, collect F , and buy back just after, at no risk. So in a frictionless market the perpetual's basis over its underlying must rise by exactly F across the settlement instant:

$$J \equiv \text{basis}(T+) - \text{basis}(T-) = F$$

This is the **ex-dividend price drop** seen from the other side. Since Elton and Gruber (1970), finance has asked whether stock prices fall by the full dividend on the ex-date, and the answer has been contested for fifty-five years, mainly because dividends and capital gains are taxed differently. Explanations range from tax clienteles (Elton & Gruber, 1970; Kalay, 1982) to microstructure frictions such as tick size and bid-ask bounce (Bali & Hite, 1998; Frank & Jagannathan, 1998). Perpetual funding is a far cleaner laboratory. The payment takes both signs. It recurs tens of thousands of times, not once a quarter. And because the payment and the price change are both profits and losses on the same derivative position, the dividend-versus-capital-gain tax wedge that dominates the equity debate is plausibly absent.

Design. Before loading any price data we pre-registered the question, the primary specification, a placebo, a natural experiment, secondary hypotheses and the interpretation of every possible outcome, and pushed the plan to a public repository (commit `377bf67`). The primary test measures the basis change across the settlement timestamp using Binance's premium index, sampled every five seconds, for the 62 USDT-margined perpetuals listed by 2020 and live through August 2026. When the results revealed a flaw in one assumption of the plan, we filed an addendum (commit `e51757a`) stating new hypotheses, and tested them only on **34 further contracts whose data had not been downloaded** when the addendum was pushed.

Findings.

1. **At the timestamp the market prices almost none of the payment.** The share of funding reflected in the basis across the settlement instant is $\beta = 0.060$. The effect is real: placebo hours give 0.003, and clock hours that become settlement times after a funding-interval change go from 0.008 to 0.220. But it is a sixteenth of the frictionless value.
2. **The instant is the wrong unit.** Binance's own documentation says settlement carries a ± 15 -second deviation and can take up to a minute to complete, so no trader can know that a position closed one second after the stamp escaped the payment. **Across the three-minute window that spans it, the basis reprices 44–45% of the funding payment**, in discovery and holdout alike.
3. **The hour before settlement moves the other way.** The basis drifts against the paying side by 42% of the payment, then reverses across the settlement window: a V-shape for positive funding and an inverted V for negative. It replicates in the holdout to the third decimal place.
4. **The underlying market moves too.** Across the window the perpetual's price moves by roughly the full payment (slope 0.95–1.14), while Binance **spot** moves by 0.44–0.66 of it. The funding schedule of a derivative moves the spot price of the asset it references.
5. **Not everything replicated.** Post-settlement drift, post-settlement order flow, a traded-price version of the instantaneous jump, and two joint hypotheses that bundled the post-settlement window with others all failed as specified. They are reported in full.

Related work. Ruan and Streltsov (2022) show that spot market quality follows a U-shape over the eight-hour funding cycle, with more volume and wider spreads near settlements. Kim and Hansen (2026) document algorithmic trading bursts at every quarter-hour mark in Binance perpetuals. Zhivkov, Todorov and Georgiev (2026) find cross-exchange funding-rate spreads peaking about two hours after settlement. He, Manela, Ross and von Wachter (2022) and Ackerer, Hugonnier and Jermann (2026)

develop the theory of perpetual pricing. To our knowledge, based on a literature search in September 2026, no study tests whether prices adjust *directionally* to the funding payment at settlement, how much of it they price, or when. The settlement-timing mechanism documented here is also, as far as we can find, new.

2. Institutional setting

The rate. Binance computes each period's funding rate as the time-weighted average premium index P of the perpetual over its index, plus an interest term: $F = P + \text{clamp}(0.01\% - P, -0.05\%, +0.05\%)$. When the premium stays within five basis points of the interest term, the clamp sets F to exactly **+1 bp** per eight hours. That happens at **47.2%** of settlements in our sample; this "baseline" payment is too small to be worth trading around. The rest spread widely: 15,110 settlements have $|F| \geq 10$ bp and 2,209 have $|F| \geq 30$ bp.

The timestamp. Settlements occur exactly on the hour: every eight hours for 93% of events, every four hours for 6%, hourly or two-hourly for the rest. Binance changed the interval of 14 contracts in our sample 32 times, mostly in 2025–2026, which provides the natural experiment in §5.1.

The window. Binance's support documentation states that "there is a 1-minute deviation in the actual funding fee transaction time," that positions within a "~15-second deviation window" of the timestamp may or may not be counted, and that "the system may require up to 1 min to complete all funding fee settlement." The payment is therefore not attached to an instant a trader can target. Our pre-registration assumed it was. That is the one substantive error in the plan, and §4 describes how it was handled.

The trap. Binance's *mark price*, which drives liquidations and unrealised profit and loss, includes a term equal to the index times one plus the last funding rate scaled by the time remaining until settlement. This term decays to zero at each settlement and resets immediately after, so the mark price carries the frictionless sawtooth *by construction*. A study that used it would find the ex-funding jump automatically. We never use the mark price as an outcome.

3. Hypotheses

The pre-registration (377bf67) fixed one primary and five secondary hypotheses.

- **H1 (primary).** At settlement, the basis jumps in the direction of funding: $J = \alpha + \beta \cdot F$, $\beta > 0$. Frictionless value $\beta = 1$. The interpretation of each outcome was fixed in advance: β significantly between 0 and 1 means **partial adjustment**, and $1 - \beta$ is the share of funding a harvester keeps.
- **H2 (placebo).** The same regression at non-settlement hour marks gives $\beta \approx 0$.
- **H3 (natural experiment).** Clock hours that become settlement times after an interval change acquire the jump.
- **H4 (arbitrage band).** The share priced rises with $|F|$.
- **H5 (order flow).** Net taker selling before positive-funding settlements, and net buying after.

- **H6 (traded prices).** The perpetual-minus-spot price gap across T gives the same answer as the premium index.

The addendum (e51757a) added five exploratory hypotheses, tested only on the holdout contracts: repricing across the settlement window (**E1**), drift against the payer in the hour before (**E2**), continued drift after (**E3**), the placebo contrast for all three windows (**E4**), and the same pattern in spot prices (**E5**).

4. Data and design

Source. All data come from Binance's public archive (`data.binance.vision`): monthly files of one-minute bars for the premium index (sampled every five seconds), perpetual trades and spot trades, plus the funding-rate history. A pipeline streams each file, extracts values around every hour mark, and discards the raw bars, which total about 20 GB.

Discovery universe (fixed before any outcome data). Every USDT-margined perpetual whose funding history begins by December 2020 and continues through August 2026: **62 contracts**. Their funding files hold 438,155 settlements. Of these, 433,856 fall at an hour mark covered by the price archive. The rest come from two contracts renamed in 2025 (EOS, MATIC), whose price archives stop while funding records continue (2,376), an archive gap for CVC in 2023 (1,918), and five before the first extracted hour. Applying the pre-registered rule, which drops events missing either bar adjacent to T , removes 2,104 more and leaves **431,752 settlements**, plus 2,893,202 non-settlement hour marks as placebos.

Holdout universe (fixed before its data were downloaded). Every USDT-margined perpetual first listed during 2021 and live through August 2026: **34 contracts**, with no overlap with the discovery set. They contribute **199,838 settlements**.

Measures. The primary outcome is the change in the premium index from its last five-second sample before T to its first sample at or after T , in basis points. The addendum's windows use one-minute bar closes:

Window	From	To
PRE: hour before	$\approx T - 60 \text{ min}$	$\approx T - 65 \text{ s}$
W: settlement window	$\approx T - 65 \text{ s}$	$\approx T + 115 \text{ s}$
POST: hour after	$\approx T + 115 \text{ s}$	$\approx T + 60 \text{ min}$

Window W begins before Binance's ± 15 -second deviation and ends after its one-minute settlement completes. It is therefore the shortest interval a trader can hold and be *sure* of being on the books at settlement.

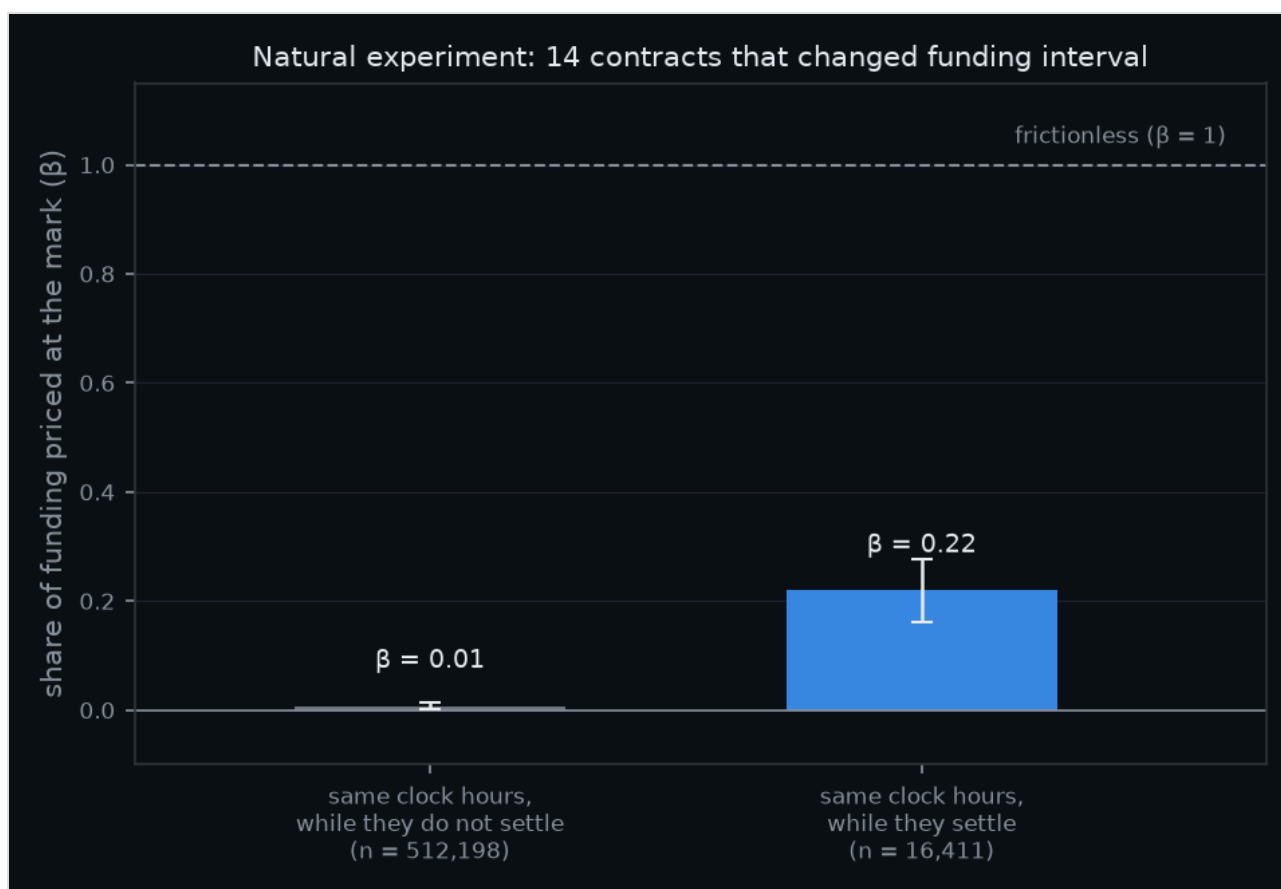
Inference. OLS with standard errors clustered by settlement timestamp, because every contract settles at the same instant: 12,505 clusters in the primary test. The secondary hypotheses are Holm-corrected as a family. A directional hypothesis whose estimate has the wrong sign is scored $p = 1$.

Timeline. (1) Pre-registration pushed at 06:54 UTC on 26 September 2026, before any price data was downloaded. (2) Pre-registered analysis run on the discovery contracts. (3) Addendum pushed at 07:15 UTC, before any holdout data was downloaded. (4) Holdout data downloaded and E1–E5 run once. Both commits are in the repository history.

5. Results

5.1 At the timestamp (pre-registered)

Test	Estimate	Result
H1 share of funding priced, β	0.060 (se 0.009, $t = 6.7$)	$\beta > 0$, $p = 2 \times 10^{-11}$
H1 against the frictionless $\beta = 1$		rejected, $t = -104.7$
H2 placebo hours	0.003 (se 0.002)	settlement – placebo = 0.057, $t = 6.2$; Holm $p = 2 \times 10^{-9}$
H3 switching clock hours, while settling	0.220 (se 0.029, $n = 16,411$)	
H3 the same hours, while not settling	0.008 (se 0.003, $n = 512,198$)	difference 0.212, $t = 7.2$; Holm $p = 2 \times 10^{-12}$
H4 β by $ F $: ≤ 1 / 1–5 / 5–10 / > 10 bp	–0.029 / 0.014 / 0.036 / 0.066	top – bottom = 0.095; Holm $p = 5 \times 10^{-5}$

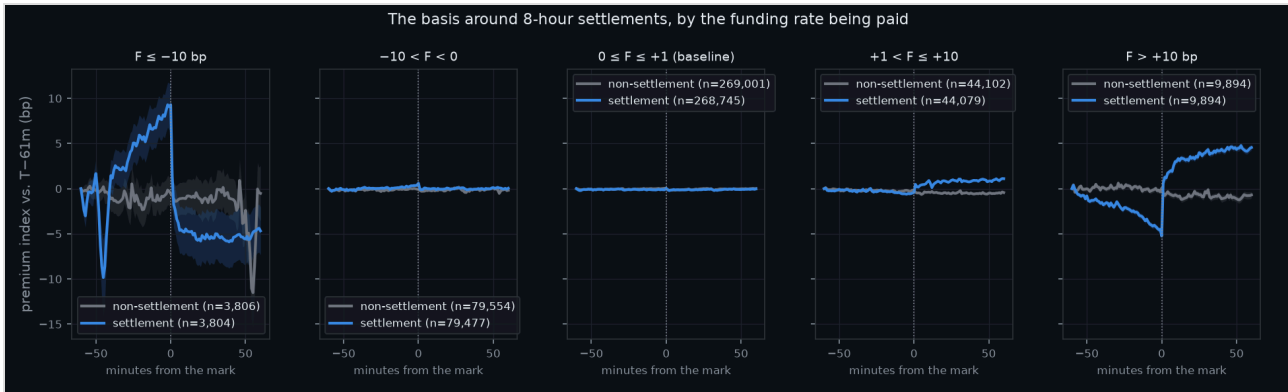


By the pre-registered rule this is **partial adjustment**, and it is robust: the estimate is significant and positive in both pre-registered periods (2020–2024: 0.027, se 0.011; 2025–2026: 0.149, se 0.017), with two-way clustering (0.060, se 0.018), with the outcome winsorised (0.015, se 0.003), and excluding baseline events (0.060, se 0.009). For BTC and ETH alone it is indistinguishable from zero (−0.022, se 0.018). The pre-registered median regression turned out to be uninformative: 24.5% of instantaneous premium changes are exactly zero (60% for the least liquid contract, 0.6% for BTC), which pins the conditional median at zero.

The effect is causal in the sense the design allows. It appears only at settlement, it scales with the payment, and it switches on at an hour of the day at the moment that hour becomes a settlement time. But it is small. Read naively, a trader who shorts one second before a positive-funding settlement and covers one second after keeps 94% of the payment.

5.2 Why the instant is the wrong unit

That trade does not exist. Binance does not settle at an instant. The payment can land up to a minute after the stamp, and a position closed within about fifteen seconds of it may or may not be charged. The event-time paths show where the repricing actually happens.

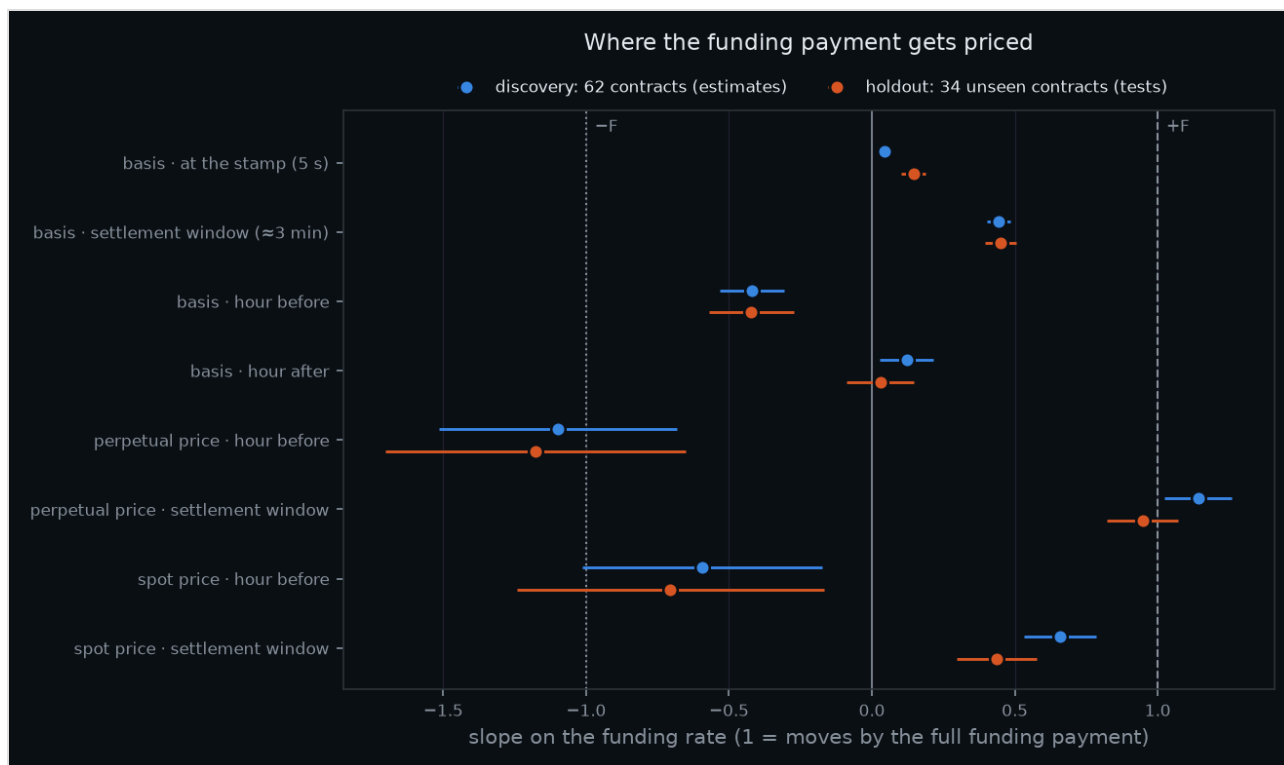


For large positive funding (right panel), the basis **falls steadily for the hour before settlement**, bottoms at the stamp, and **jumps in the minute after**. Large negative funding (left panel) is the mirror image. Non-settlement hours are flat in every panel, and so are settlements at the baseline rate. The five-second measurement sits exactly on the turning point of this V, which is why it records almost nothing.

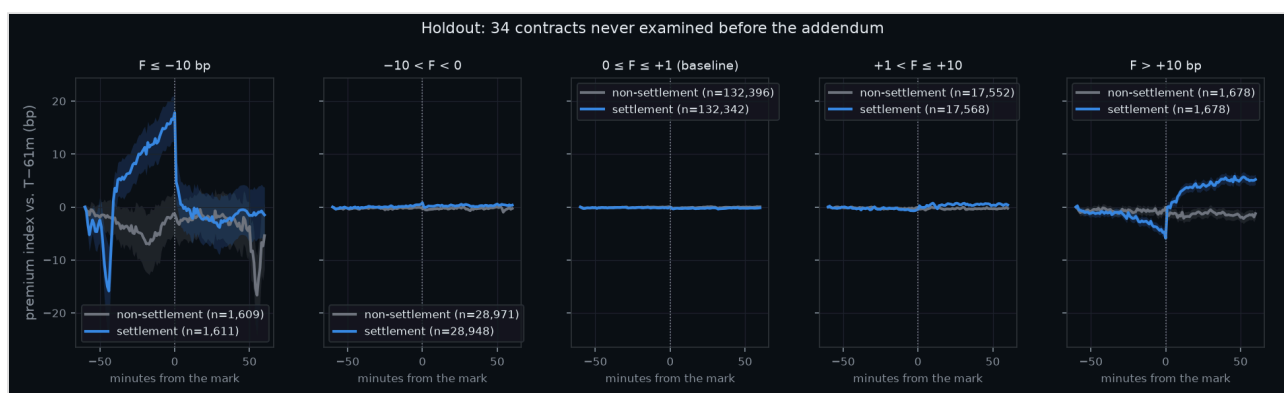
The pre-registration had not anticipated this. The addendum therefore fixed the window W , defined as the shortest hold that guarantees being on the books, and tested the hypotheses it implied on contracts not yet downloaded.

5.3 Across the settlement window (addendum; tested on the holdout)

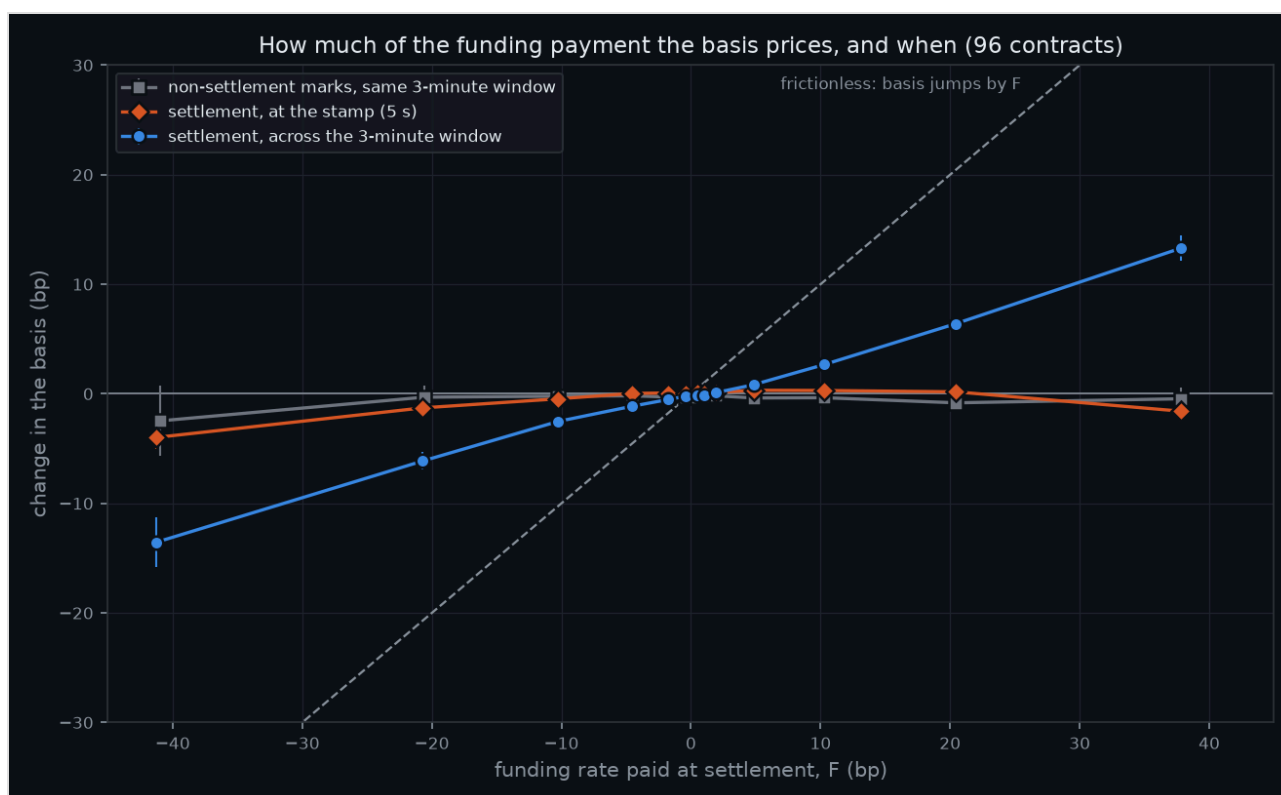
	Discovery: 62 contracts (estimates)	Holdout: 34 unseen contracts (tests)	Holm p (holdout)
E1 basis across window W	0.444 ($t = 21.0$)	0.449 ($t = 16.1$)	3×10^{-18}
E1 minus the instantaneous β	0.398 ($t = 19.3$)	0.303 ($t = 8.9$)	
E2 basis, hour before	-0.420 ($t = -7.3$)	-0.422 ($t = -5.6$)	1×10^{-7}
E3 basis, hour after	0.122 ($t = 2.5$)	0.029 ($t = 0.5$)	1: not confirmed
E4 settlement – placebo: W / PRE / POST	0.441 / -0.420 / 0.090	0.392 / -0.446 / -0.128	1: not confirmed (POST)
E5 spot: PRE / W / POST	-0.59 / 0.66 / -0.11	-0.71 / 0.44 / -0.19	1: not confirmed (POST)



The two central quantities replicate almost exactly: 0.444 becomes 0.449, and -0.420 becomes -0.422 , on contracts that played no part in forming the hypotheses. The holdout paths show the same V:



Every prediction about the hour *after* settlement failed. The discovery paths suggested continued drift in the direction of funding; the holdout shows none. E4 and E5 fail as filed because each bundled a post-settlement prediction with two others. Their other components hold in the holdout (spot PRE -0.71 , $t = -2.6$; spot W 0.44 , $t = 6.1$), but that is a statement about sub-components, not a confirmation.



5.4 The underlying market moves too

The basis is the perpetual's price relative to the index. Decomposed, **the perpetual's own price reprices by about the full payment across the window** (slope 1.14 in discovery, 0.95 in the holdout) after falling by about the full payment in the hour before (-1.10 , -1.18). Binance **spot** carries roughly half of each move: 0.66 and 0.44 across the window, -0.59 and -0.71 before. The spot market pays no funding and has no settlement. Its price moves because a derivative written on it has a payment schedule.

5.5 Robustness of the window results (exploratory)

- **Breadth.** β_W is positive in 62 of 62 discovery contracts and 32 of 34 holdout contracts. β_{PRE} is negative in 60 of 62 and 32 of 34. Both keep their sign in every calendar year of both samples.
- **No look-ahead.** Replacing F with the *previous* settlement's rate, known eight hours in advance (correlation 0.74 and 0.65 with F), leaves β_W at 0.36 ($t = 17.8$) and 0.33 ($t = 13.0$).
- **Tails.** Winsorising F changes little (β_W 0.44 and 0.51). Winsorising the *outcome* at 1% shrinks β_W to 0.18 and 0.16, so a large part of the magnitude comes from large moves during extreme-funding episodes. The sign does not depend on them.
- **One non-result.** The perpetual's *price* decline before settlement is not predictable from the lagged rate (-0.19 , $t = -0.6$; holdout 0.03). The decline in the *basis* is (-0.29 , $t = -5.7$; holdout -0.17 , $t = -4.0$). Only the relative move can be forecast in advance.

5.6 Mechanism: order flow

Taker order flow in the fifteen minutes before settlement is strongly decreasing in F : slope -0.00174 per bp, $t = -17.7$. More funding means more aggressive selling before a positive-funding settlement, as

payers exit and receivers position. That half of H5 holds. The other half, net buying after settlement, does not: the slope is -0.00030 ($t = -4.4$), the wrong sign, so H5 fails as specified. The re-entry that reverses the price does not arrive as aggressive taker buying in the next fifteen minutes. It may arrive through resting limit orders, or within the minute the bars cannot resolve.

H6 also fails as specified. Across the five-second instant, the perpetual-minus-spot traded-price gap does not load on F (-0.007 , $t = -1.8$), because perpetual and spot move *together* (0.039 and 0.045, both $t > 14$). That was the first sign of \$5.4, visible before the addendum was written.

6. What a harvester keeps

The economically relevant quantity is the payment a trader can collect while holding across the whole window: $|F| - \text{sign}(F) \cdot \Delta W$, per settlement, before costs.

$ F $	Settlements (disc. / hold.)	Mean $ F $	Kept, index hedge	Kept, Binance spot hedge	Net of 8 bp (spot hedge)
≤ 1 bp	326,718 / 159,179	0.8 bp	108% / 120%	0.7 / 0.6 bp	-7.3 / -7.4 bp
1–5 bp	67,210 / 29,156	2.5 bp	81% / 78%	1.8 / 1.9 bp	-6.2 / -6.1 bp
5–10 bp	20,278 / 6,971	6.9 bp	80% / 77%	5.0 / 5.2 bp	-3.0 / -2.8 bp
> 10 bp	14,606 / 3,758	21.8 / 26.2 bp	65% / 65%	12.9 / 14.9 bp	+4.9 / +6.9 bp

The index hedge is not tradeable; Binance's index is a composite of several exchanges. The spot hedge is, and costs a round trip in each market, 8 bp in total at the base maker rates. On that basis, **funding payments below 10 bp are not worth harvesting, and payments above it keep about 56–59%**, a net 5–7 bp per event in both samples. Three caveats apply. These large-funding settlements are about 3% of the sample and concentrate in smaller contracts during speculative episodes, when depth is thin, so capacity is limited. The funding rate is treated as known 65 seconds before settlement; under Binance's time-weighted formula the last minute carries well under 1% of the weight, so this is close to, but not exactly, what a trader sees. And the ± 0.2 –0.5 bp intervals in the table are event-level, not clustered.

7. Discussion

Why the instant prices nothing. The frictionless argument needs a moment before which a position pays and after which it does not. Binance provides a blurred one. A trader cannot safely re-enter until the settlement is certainly over, so the repricing happens inside a window in which no one can be sure of their exposure. The result is a V: the basis moves against the payer as positions are cut or harvesting

shorts are opened before the window, then reverses within it. Neither leg can be traded at the stamp itself.

A limit to arbitrage made by the exchange. Settlement-timing uncertainty is a design choice, not a fundamental friction, and it is enough to leave 56–59% of large payments unpriced across the window, which is what a hedged trader holding through it collects. The ex-dividend literature explains incomplete price drops with taxes or tick size. Here neither is present. The residual comes from not knowing exactly *when* the payment happens.

The ex-dividend ratio has a time dimension. Measured at the stamp, the "price-drop ratio" of funding is 0.06; across three minutes, 0.45 on the basis and about 1 on the perpetual's own price. Which number is "the" adjustment depends on which instant the payment is attached to. That question could be asked of equity ex-dates too, where the payment's legal timing (record date, ex-date, pay date) is also spread out.

Derivative design moves the underlying. About half of the repricing appears in Binance spot. The mechanism is presumably arbitrage between the two, carrying funding-motivated order flow from the perpetual into the underlying. It means a purely contractual feature of a derivative, its payment calendar, predictably moves the spot price of the asset, three times a day.

The effect is growing. At the stamp, β rose from 0.027 in 2020–2024 to 0.149 in 2025–2026, and in the holdout, whose contracts are younger and less liquid, it is 0.146. More of the payment is being priced at the stamp over time, though still a small fraction.

8. Limitations

One venue: Bybit and OKX settle at the same instants, and cross-venue replication is the natural next step. One-minute bars cannot say where inside the settlement minute the repricing lands; trade- or quote-level data could. The premium index is Binance's construction and its index is not tradeable, which is why §6 reports a spot hedge. Fees vary by account tier, so 8 bp is a base-rate illustration. A large part of the window effect's magnitude comes from extreme-funding episodes, although its sign does not. The post-settlement claims failed and are not claimed. The discovery and holdout samples come from the same exchange and period, so they are independent in contracts, not in market-wide shocks. And the pre-registration's assumption of an instantaneous settlement was wrong. The addendum was the response, and the paper reports both.

9. Conclusion

A perpetual future's funding payment is a known, scheduled cash flow, and at the instant it is paid the market prices about 6% of it. That is not a failure of arbitrage in the usual sense, because there is no such instant: Binance settles over a blurred window of up to a minute. Measured across the window, the basis reprices about 45% of the payment after drifting 42% the other way in the hour before, the perpetual's own price reprices by roughly all of it, and spot moves by half. All of this held, almost to the third decimal place, in contracts that were not examined until the hypotheses had been filed. What remains is

a small, capacity-limited residual for large payments, and a design lesson: a payment's price is attached to the moment traders can be certain it happened.

Deviations from pre-registration

1. **Settlement timing.** The plan assumed an instantaneous settlement. Binance's documentation, consulted after the event-time paths were seen, describes a window of up to a minute. The pre-registered results are reported as filed. The window analysis was filed as an addendum before the holdout was downloaded, and is tested only there.
2. **Operationalisations the plan left open.** For H4, the Holm p -value tests the top-bin slope against the bottom-bin slope. For H5, both predictions must hold, so its p -value is the larger of the two. For H6, the test is the traded-price slope.
3. **Median regression.** Uninformative because 24.5% of outcomes are exactly zero. Reported, not interpreted.
4. **One event seen early.** While validating the month-boundary logic of the data pipeline, one settlement's premium values (BTCUSDT, 2024-03-01 00:00 UTC) were printed, before the plan's analysis ran. That is one of 431,752 events.
5. **Reproducibility changes that do not affect estimates.** A funding-file download step and retry logic were added to the pipeline; the display intervals in the natural-experiment figure use each regime's own clustered standard error. The pre-registered test statistics were re-run after these changes and matched to every printed digit.

References

- Ackerer, D., Hugonnier, J. & Jermann, U. (2026). *Perpetual Futures Pricing*. Mathematical Finance.
- Bali, R. & Hite, G. (1998). *Ex-dividend day stock price behavior: discreteness or tax-induced clientele?* Journal of Financial Economics.
- Binance. *Introduction to Binance Futures Funding Rates* (support FAQ).
- Elton, E. & Gruber, M. (1970). *Marginal stockholder tax rates and the clientele effect*. Review of Economics and Statistics.
- Frank, M. & Jagannathan, R. (1998). *Why do stock prices drop by less than the value of the dividend? Evidence from a country without taxes*. Journal of Financial Economics.
- He, S., Manela, A., Ross, O. & von Wachter, V. (2022). *Fundamentals of Perpetual Futures*. Working paper.
- Kalay, A. (1982). *The ex-dividend day behavior of stock prices: a re-examination of the clientele effect*. Journal of Finance.
- Kim, C. & Hansen, P. R. (2026). *The Quarter-Hour Effect: Periodic Algorithmic Trading and Return Predictability in Cryptocurrency Futures*. Working paper.
- Ruan, Q. & Streltsov, A. (2022). *Perpetual Futures Contracts and Cryptocurrency Market Quality*. Working paper.

- Zhivkov, P., Todorov, V. & Georgiev, S. (2026). *Temporal Dynamics of Market Microstructure in Cryptocurrency Perpetual Futures*. International Journal of Financial Studies.
 - Martingale, Research Notes 001–006.
-

How to cite

Gilani, N. (2026). *Ex-Funding Days: How Perpetual Futures Price a Scheduled Payment*. Martingale, Research Note 007. <https://github.com/hilothefunnydog123-coder/quant-research>

© 2026 Neil Gilani. Code: MIT License. Text, figures, and findings: CC BY 4.0 (reuse with attribution).